

Environmental Status at Indian Antarctic Station Maitri

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INTRODUCTION

Monitoring and analysis of water from Priyadarshini Lake, at Indian Antarctic Station - Maitri, which is a source of drinking water for Maitri.

Performance evaluation of wastewater treatment plant: Based on the analysis of influent and effluent, performance of the treatment plant may be evaluated. This will confirm whether the effluent quality is within the permissible limit set by the pollution regulatory authorities.

SCOPE OF STUDY DURING XXII IAE

Period of study: Summer expedition January 2003 to March 2003

- Monitoring of water source to Maitri (Lake waters) and surface waters feeding to lakes (Tributaries)
- Analysis of lake waters
- Analysis of waste water generated at Maitri
- Performance evaluation of treatment plant

WORK PLAN DURING XXII IAE

- Setting up of Laboratory
- Data collection
- Collection and analysis of samples
- Inferences regarding lake water quality
- Performance evaluation of treatment plant

To carry out the performance study a temporary laboratory was set up at Aravali summer hut, Maitri Station. V.J.T.I. Matunga, Mumbai 400019, gave equipment and chemicals required for experimentation.

Following sequence of works were undertaken during XXII IAE.

- Monitoring program of Priydarshini lake water as well as some other surface waters as control waters for the quality comparisons
- Performance evaluation of the wastewater treatment facilities at Maitri
- Monitoring of noise levels at various locations in and around station.
- Study about the solid waste management
- Study of the Soil samples collected from the adjacent areas to Maitri.

WATER

i) Fresh Water

Priydarshini lake is a source of water for the Maitri station from its inception. It's more than a decade that the water is used for all the purposes at Maitri. Water samples were drawn from all around the Priydarshini lake and from the various locations from the middle of the lake. Water samples were also drawn from glaciers and a lake about 25 Km from the Maitri station. These samples were taken and analyzed from the point of view of control samples for comparisons. A few water samples were also analyzed at the Institute's laboratory to have inter laboratory studies and to have better control over the quality of analysis of samples.

Water samples were collected from 26 various locations (Ref. Table 1). The samples were characterized for various parameters as physical, chemical and biological. These included few field parameters as pH, temperature, conductivity, turbidity and dissolved oxygen (D.O), chloride, sulphate, bacterial contamination and a few heavy metals.

Table 1 Sampling location for water samples

Sample No.	Location	G.P.S. Coordinates
1	Raw lake water	70°45'53" 11°43'52.6"
2	Tap water 'Kitchen'	—
3	Glacier water 'Behind station'	—
4	Lake (Pump House)	—
5	Lake water	70°45'53" 11°43'52.6"
6	Lake water	70°45'52.9" 11°43'43.8"
7	Lake water	70°45'55.4" 11°43'40.6"
8	Lake water	70°45'52.2" 11°43'24.5"
9	Lake water	70°45'47.3" 11°43'32.5"
10	Lake water	70°45'44.5" 11°43'58.4"
11	Lake water	70°45'40.1" 11°43'34.1"
12	Lake water	70°45'39.2" 11°43'09.1"
13	Lake water	70°45'44.4" 11°43'54.7"
14	Lake water	70°45'46.6" 11°43'47"
15	Lake water	70°45'0.6" 11°43'13.8"
16	Lake water	70°45'03.4" 11°43'20.9"
17	Lake water	70°45'58.1" 11°43'46.6"
18	Lake water	70°45'54.3" 11°43'39.5"
19	Lake water	70°45'50.6" 11°43'50.6"
20	Lake water	Middle of lake
21	Lake water	Middle of lake
22	Lake water	Middle of lake
23	Lake water	Middle of lake
24	Lake water	Middle of lake
25	Control lake 1	70°45'53" U°43'52.6"
26	Control lake 2	70°45'53" 11°43'52.6"

ii) Wastewater

The wastewater is generated at Maitri station from various day-to-day activities of the occupants from kitchen, bathroom, washrooms and urinals. This wastewater is disposed in to the Priyadarshini lake after the

treatment at the Maitri station. The treatment provided to the wastewater is mainly by passing through the bio-discs (Two in number, separately provided for two wastewater streams (i) Kitchen and (ii) Bathrooms & Urinals). Later on the effluent of the bio discs is collected in a open pond for minimum 3 weeks in summers (December to February). The wastewater after passing through pond reaches ultimately to the Priydarshini lake.

The composite samples were collected from various locations and at various times for the analysis. Total of 13 samples were analyzed for characterization. Table 2 shows the various locations from where the wastewater samples collected for the study.

Table 2 Wastewater sampling locations

Sample No.	Location
1	Urinals and bath effluent
2	Kitchen waste water effluent
3	Urinals and bath effluent
4	Kitchen waste water effluent
5	Kitchen waste water influent
6	Kitchen waste water effluent
7	Urinals and bath influent
8	Urinals and bath effluent
9	Kitchen waste water influent
10	Kitchen waste water effluent
11	Urinals and bath influent
12	Urinals and bath effluent
13	Waste stabilization pond

The following parameters were observed during the study

1. The COD analysis of the influent to and effluent from the biodiscs to study the performance evaluation of the wastewater treatment.
2. The wastewater discharge estimations.

The effluents of the biodiscs B_R (biodiscs kitchen) & biodiscs urinal & bath B_{UB} (biodiscs urinal & bath) were fluctuating throughout the day. From the peak and lean discharges for B_K and B_{UB} . There is negligible discharge during 2300 Hrs to 0600 Hrs as all the pumps (Both supply & delivery) are shut down. Therefore, the effective supply to the biodiscs is only for 17 hours.

The discharge measured at the peak and lean periods for both the biodiscs were averaged out to get the total volume of wastewater generated in liters per day (LPD).

To Biodiscs	Period / Time slot	Quantity of wastewater	Time of collection	Approximate Daily Wastewater discharge : LPD
A)B _K B _{UB}	Peak	10 liters	03 Minutes	17 Hrsx60 Min.x(10/3.5) = 2914
	Lean	09 liters	21 Minutes	17 Hrsx60 Min.x(09/26) = 353
B)B _K B _{LB}	Lean	10 liters	4 Minutes	17 Hrsx60 Min.x(10/04) = 2550
	Peak	10 liters	30 Minutes	17 Hrsx60 Min.x(10/30) = 340
Average daily wastewater discharge = (3267 + 2890)/2 = 3079 s 3080 LPD				

The average demand for consumption of fresh water is approximately 3500 LPD. Continuous monitoring of the discharge cannot be carried out without adequate arrangements at the Maitri station Antarctica.

Effluent from the biodiscs is collected in a pond and later after few days it is pumped in to the lake. The periodicity of pumping depends upon the weather.

It was noted during the study that the discharge of treated effluent is in the same lake from where the water is drawn for the Maitri. The water from the summer camp facilities used during the summer also seeps to the same lake. All the residential facilities are on higher elevation and the ground /soil conditions are rocky. There is very little of retention of the water from the wastewater pond. All the treated wastewater seeps through rocky bed to the lake. This action of seeping also provides some treatment.

iii) Human waste (Excreta) and other organic, inorganic solid waste disposal

Incinerating the waste carries out Excreta/human waste disposal. The night soil is collected separately without mixing it with the liquid part of waste as urine / ablution water. The users are abstained from urinating and using ablution waters. The waste is directly collected in a chamber where it is incinerated with the help of burners using aviation turbine fuel (ATF). After the complete burning of the organic waste i.e. till it is converted to ash. The ash is later on collected in plastic bags and packed for departing from the Antarctic main land.

The following observations were made during the study of the human waste disposal system:

- ^* The arrangements for the collection and disposal of the sullage (liquid waste from washing places and urinals) generated at summer camp are not proper. The collection system needs proper

maintenance and disposal in a scientific manner. At present it is directly drained to the open area adjoining the camp. The quantity of the sullage generated is relatively meagre and disposal is on a large area, climate during the period of summer camp is also conducive towards safe disposal. But it adds to the unsightly conditions. There were no signs of the sullage reaching to the nearest point on lake.

- > The toilets with incinerators are working properly and the objective of safe disposal of night soil is achieved. Only the time required for burning / thermal treatment was more and could have been reduced by providing the multi burner arrangement. Which will reduce the burning time as well the hardships of the gaily duty personnel.
- > The solid waste generated in the kitchen is mostly organic in nature. The organic wastes are peelings, food wastes and paper waste. This waste is also burnt in a incinerator and the ash is collected and packed for dispatching away from Antarctic land.

iv) Soil

Soil samples were collected for its characterization from areas close to Maitri and Priyadarshini lake for studying the impact of wastewater pollution from Maitri.

SAMPLING

Representative samples of water were collected from the Priyadarshini lake from all the locations as well from the center of lake also. The glacial water sample and a distant lake sample were collected to have characteristics of water from locations away from the human activities. These water samples were collected in order to act as the controls.

RESULTS / OBSERVATIONS

Water samples collected from various locations were characterized for various parameters like physical, inorganic and organic parameters. These include pH, temperature, conductivity, turbidity, alkalinity, total hardness, chloride, sulphate, nitrate, nitrite, dissolved oxygen, chemical oxygen demand (COD), biological oxygen demand (BOD), some heavy metals. The analysis results are summarized in Tables from 3 to 8.

Tables from 3 to 5 show the various observations for Fresh water and Tables from 6 to 8 for that of Wastewater.

Table 3 Water analysis - Physical Parameters

Sample No.	Color	Temp. Sample °C	Odor	PH	Mv	Temp. Test °C
1	clear	4	odorless	7.21	—	20.00
2	clear	4	odorless	—	—	.
3	clear	5	odorless	5.31	—	—
4	clear	2	odorless	6.3	30.0	—
5	clear	4	odorless	6.34	40.3	15.80
6	clear	2	odorless	6.05	52.0	13.90
7	clear	5	odorless	6.07	54.0	14.00
8	clear	5	odorless	6.34	34.0	14.00
9	clear	5	odorless	6.21	45.8	16.00
10	clear	3	odorless	6.03	53.6	14.20
11	clear	3	odorless	6.22	44.5	16.20
12	clear	3	odorless	6.09	51.1	14.50
13	clear	6	odorless	6.20	45.2	18.90
14	clear	6	odorless	6.16	45.8	11.00
15	clear	6	odorless	6.58	22.9	11.80
16	clear	4	odorless	6.77	13.3	11.40
17	clear	4	odorless	6.64	19.3	11.80
18	clear	4	odorless	6.59	22.5	07.90
19	clear	4	odorless	6.58	23.4	08.40
20	clear	5	odorless	6.61	24.7	07.30
21	clear	5	odorless	6.45	30.7	09.40
22	clear	5	odorless	6.30	37.6	09.40
23	clear	6	odorless	6.20	45.4	09.20
24	clear	6	odorless	6.23	43.9	14.70
25	clear	6	odorless	6.96	2.2	18.30
26	clear	6	odorless	7.06	1.5	12.90

Table 4 Water analysis - Chemical Parameters

Sample No.	Chloride mg/L	Nitrates mg/L	Sulphites mg/L	D.O.(Dissolved Oxygen) mg/L
1	6	35	4	7.50
2	6	50	3.5	7.35
3	20	100	3	
4	14	50	3	7.35
5	8	50	2	6.80
6	7	65	2.5	6.80
7	18	50	4.5	9.30
8	14	100	4.0	7.40
9	14	100	2.0	8.05
10	10	50	2.5	6.65
11	10	65	3.0	6.60
12	18	75	3.5	6.80
13	16	50	3.5	6.20
14	18	70	3.5	6.00
15	10	75	2.0	6.90
16	10	70	2.0	6.45
17	8	65	2.0	6.70
18	10	75	3.0	6.80
19	8	60	2.0	7.80
20	12	70	3.0	6.30
21	8	75	3.0	6.30
22	10	50	2.0	6.45
23	12	70	2.0	6.15
24	10	70	2.0	6.15
25	200	15	5.0	7.86
26	200	20	4.0	7.00

Table 5 Water analysis - Chemical & Biological Parameters

Sample No.	Iron mg/L	Manganese mg/L	Aluminium mg/L	Presence of Bacterial contamination
1	0.10	0	0	Nil
2	0.10	0	0	Nil
3	0.25	0	0	Nil
4	0.10	0	0	Nil
5	0.15	0	0	Nil
6	0.10	0	0	Nil
7	0.20	0	0	Nil
8	0.15	0	0	Nil
9	0.20	0	0	Nil
10	0.10	0	0	Nil
11	0.10	0	0	Nil
12	0.15	0	0	Nil
13	0.50	0	0	Nil
14	0.20	0	0	Nil
15	0.10	0	0	Nil
16	0.10	0	0	Nil
17	0.10	0	0	Nil
18	0.10	0	0	Nil
19	0.15	0	0	Nil
20	0.05	0	0	Nil
21	0.05	0	0	Nil
22	0.05	0	0	Nil
23	0.05	0	0	Nil
24	0.05	0	0	Nil
25	0.20	0	0	Nil
26	0.10	0	0	Nil

Table 6 Wastewater analysis - Physical Parameters

Sample No.	Location	pH	Mv	Test Temp. -C	Total solids mg/L
1	Urinals and bath effluent	6.75	.	-	-
2	Kitchen waste water effluent	4.96	114.0	-	-
3	Urinals and bath effluent	7.24	14.0	14.0	-
4	Kitchen waste water effluent	5.39	91.1	13.9	-
5	Kitchen waste water influent	4.97	116.3	22.0	2400
6	Kitchen waste water effluent	6.04	53.2	17.8	400
7	Urinals and bath influent	8.45	84.1	23.3	2800
8	Urinals and bath effluent	7.58	33.2	18.6	600
9	Kitchen waste water influent	5.73	71.3	16.8	1650
10	Kitchen waste water effluent	5.57	80.1	18.8	500
11	Urinals and bath influent	6.98	1.1	11.6	400
12	Urinals and bath effluent	7.17	-36.6	15.6	800
13	Waste stabilization pond	8.96	107.7	10.7	550

Table 7 Wastewater analysis - Physical Parameters

Sample No.	Location	Chloride mg/L	Nitrates mg/L	Sulphites mg/L
1	Urinals and bath effluent	.	> 150	240
2	Kitchen waste water effluent	-	-	65
3	Urinals and bath effluent	1460	125	7
4	Kitchen waste water effluent	600	50	8.5
5	Kitchen waste water influent	98	50	7
6	Kitchen waste water effluent	80	50	9
7	Urinals and bath influent	130	40	17
8	Urinals and bath effluent	850	100	400
9	Kitchen waste water influent	18	50	5.5
10	Kitchen waste water effluent	50	35	2.5
11	Urinals and bath influent	700	150	500
12	Urinals and bath effluent	60	125	700
13	Waste stabilization pond	250	125	50

Table 8 Wastewater analysis - Chemical Parameters

Sample No.	Location	HS mg/L	Chemical Oxygen Demand mg/L	D.O. mg/L
1	Urinals and bath effluent	0.02	-	-
2	Kitchen waste water effluent	0.00	-	-
3	Urinals and bath effluent	0.02	-	7.0
4	Kitchen waste water effluent	0.02	.	1.0
5	Kitchen waste water influent	0.00	800	-
6	Kitchen waste water effluent	0.00	400	.
7	Urinals and bath influent	0.13	600	-
8	Urinals and bath effluent	0.01	240	.
9	Kitchen waste water influent	0.00	520	-
10	Kitchen waste water effluent	0.00	400	-
11	Urinals and bath influent	0.04	300	-
12	Urinals and bath effluent	0.02	288	.
13	Waste stabilization pond	0.04	540	-

ADDITIONAL OBSERVATIONS

Noise levels at various places at different time were also recorded during the study. The average noise levels are presented in Table 9.

Table 9 Noise level measurements

S. No.	Location / Source	Noise levels (dBA)
1	Generator I - Room / container entry	98
2	Generator I - Room / container entry	98
3	Incinerator room	80
4	In open between generator complex & Maitri	70
5	Boiler room	74
6	Noise due to wind with wind velocity of 22 knots	75
7	Maitri Lounge at noon	61
8	During midnight no winds at Aravali summer hut	25

Table No. 10 shows the results of the analysis conducted on Surface water / Lake water samples.

**Table 10 Surface water / Lake water samples analysis
in the VJTI laboratory**

Parameter	Sample I	Sample II	Sample III
Color	colorless	colorless	colorless
Temperature Sample °C	28	28	27
Odor	odorless	odorless	odorless
pH	6.5	6.8	6.6
Test Temperature °C	28	28	27
Turbidity in N.T.U.	0.60	0.40	0.50
Total Hardness mg / L	62.00	70.00	60.00
Chlorides mg / L	5.00	5.0	6.00
Sulphates mg / L	6.00	5.00	6.00
Nitrate mg / L	15.00	12.00	10.00
M.P.N, index per 100 ml	Nil	Nil	Nil

CONCLUSIONS

1. The quality of water from Priyadarshini lake, which is used as a source of water for Maitri is fit for human consumption / drinking.
2. The wastewater / sullage generated at Maitri station is treated with the biodiscs and later put in the waste stabilization pond. The performance evaluation of the treatment was analyzed separately at various stages. The efficiency of the biodiscs was approximately 40 %. Further the effluent of the Klogestures is stored in the open waste stabilization pond. During summers an appreciable degree of treatment is achieved. Later on, as per the suitability of weather the treated wastewater is pumped in to the Lake. During this the water passes through the soil / rocky bed along with the seepage from the pond. During seepage of waste water and pumping filtration takes place and further refinement of the wastewater is achieved. Lastly and most importantly, the volume of wastewater generated ranges approximately at a rate of 2600 LPD to 3000 LPD dilutes in a water body of (0.45 sq. km surface area with an average depth of 3 meters) 1350000 cubic

meters. All these parameters are responsible for practically negligible impact on the lake water.

The efficiency of the treatment provided to waste water is less but when aggregated at the disposal point it worked out to be and not affecting the water in the Priyadarshini lake.

3. The study of the human waste disposal system and the kitchen solid waste was found to work satisfactorily without affecting the nearby atmosphere so far.
4. The noise levels found inside the generator and incinerator rooms are higher and advise for the regular maintenance of machine parts and use of earplugs for the operators, to reduce the effect of noise levels.
5. Analysis of the soil samples collected from Maitri is yet to be received in the laboratory for analysis and confirmation of the impact of the waster water results.

RECOMMENDATIONS / SUGGESTIONS

1. Biodiscs were found to work at much lower temperature than the expected optimum temperature of 25°C to 30°C. That is one of the reasons for the low efficiency of he biodiscs. For any bacterial growth and action temperature is important. It is recommended to have the biodisc units inside the main station to maintain the temperatures.
2. The arrangements for the proper collection and safe disposal of the wastewater from the summer camp washrooms need to be designed, constructed and maintained properly.
3. The ash from the incinerators and the toilets after burning the human waste can be disposed on the oasis main land as it is total inorganic ash. For this the conditions laid down under international protocol needs to be referred.

RECOMMENDATIONS FOR FUTURE WORK

1. An environmental monitoring laboratory needs to be set up for monitoring air, water and solid wastes generated and disposed safely. This laboratory will also report about the international protocol about the waste on the Indian Antarctic Station and its impact on the near environment.
2. The study about the heavy metals such as mercury, bad and other organic pollutants in the environmental matrix of Antarctica. This will necessitate the sophisticated instruments in the laboratory.

ACKNOWLEDGEMENT

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MID-WINTER DAY CELEBRATIONS

Games competitions.....

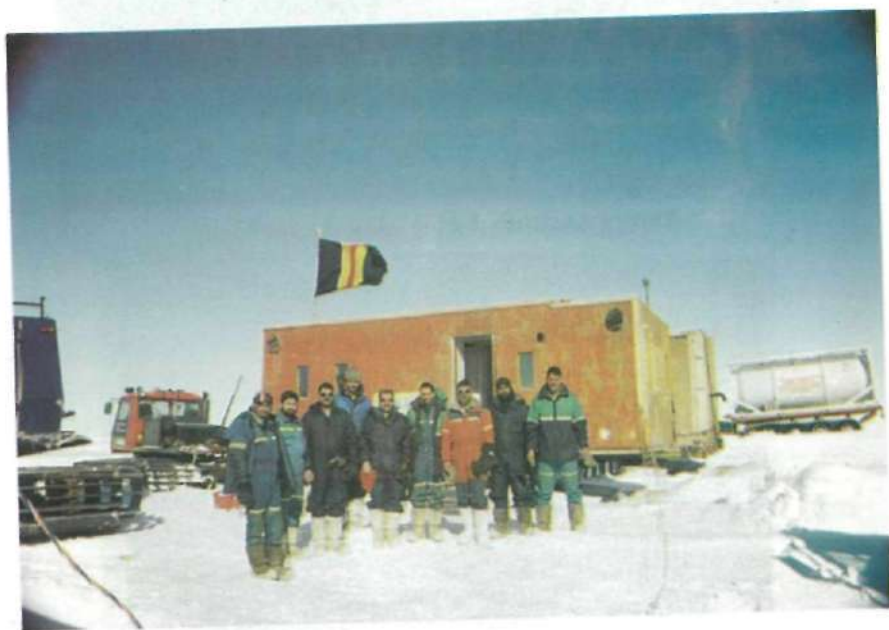


Indo-Russian Get-
to-gather, Music
and Party.....





Appendicectomy Operation in Progress



Convoy Team During Convoy Operation at Dakshin Gangotri in the Background is the Banjara Living Module



Last Sun Good Bye for Long Rest



Aurora Australis during long Winter Night



First Sun after two and Half Months Yesiam here again for the Service



Convoy Caught in 7 Days Bizzard



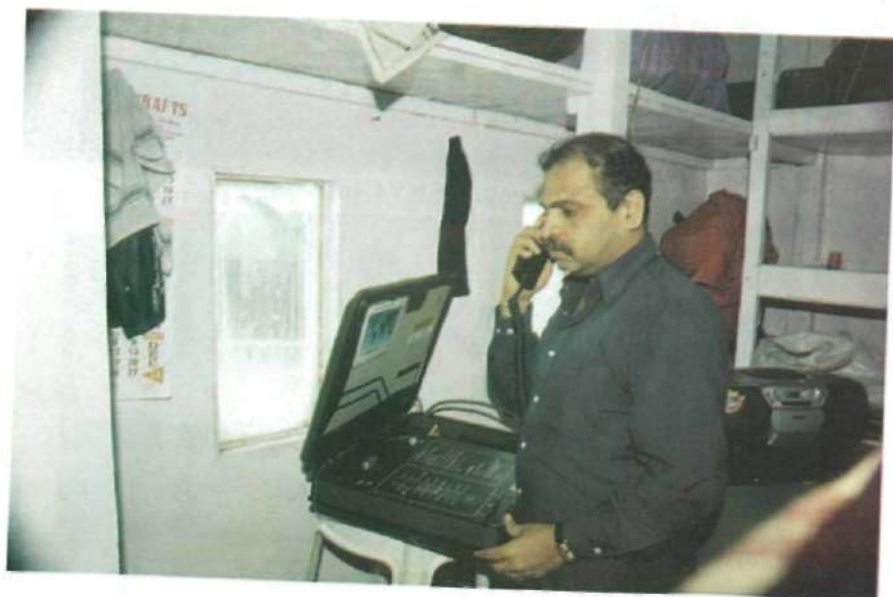
Fitting Engine to 125 KV Generator



Fixing Hut for NGRI Experiments



Installation of Relay Station at VATAIYHA



We are fine need not to worry Using mobile sat-com at DG to convey well being of convoy members